3W Well Data Viewer

A desktop GUI to inspect the real instances of the 3W dataset

Thiago Tomás de Paula

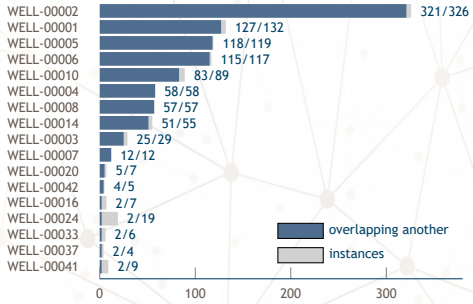
University of Brasília

5th Workshop on the 3W dataset
November 2026

Overlapping samples

Timelines page ► How are the instances of one well distributed in time?

17(42.5%) wells have overlaps, totalling 986(88.11%) instances



Example: instances overlapping in WELL-00004.



Worries:

- Overlapping samples may land in **train and test**, causing leakage.
- Duplicated stretches pull **clusters** tighter, inflating scores.

Thankfully, 3W 2.0.0 has no label collisions, making joins principled and avoiding leakage.

Clustering must still be carefully calculated!

What a joined history is labeled as

Timelines page ► How are the instances of one well distributed in time?

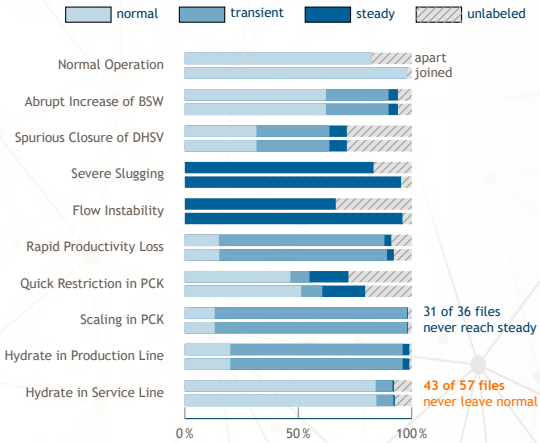
12.3 % ▷ 2.9 %
of unlabels after join

After the join, most unlabeled stretches appear trivial and might be recoverable.

Unlabeled stretch position	*n*	Duration	Share
head of a recording	208	1 h each	86.5 %
seam between equal labels	882	median 76 s, max 56 min	11.5 %
between two labels	16	median 6 min, max 1 h	2.0 %

▸ Appendix

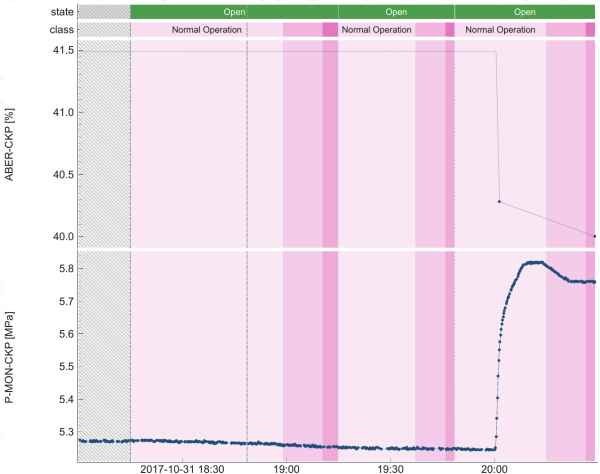
Each fault folder's recorded time, by label, its files apart and joined



Histories whose labels do not add up

Timelines page ► How are the instances of one well distributed in time?

WELL-00004's joined Quick Restriction in PCK instances.



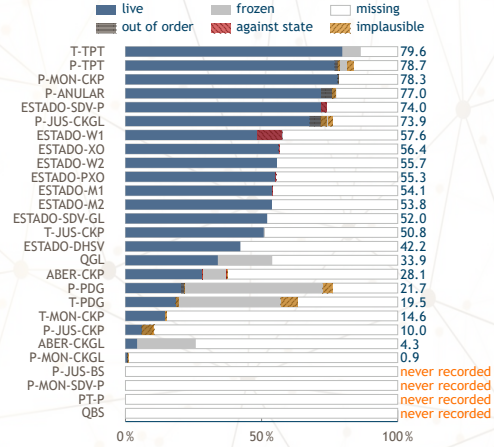
- Joined histories go back to normal **13 times** in 7 wells, **every time at a file boundary**.
- **WELL-00004**, left: three quick restrictions in 75 min, each back to normal 5 to 15 s after its steady state.
- **WELL-00006**: Abrupt Increase of BSW, steady, then normal 30 min later in a Scaling in PCK file.
- The others: Flow Instability (WELL-00001, 00004), Rapid Productivity Loss (00016, 00020), Scaling in PCK (00024), Hydrate in Service Line (00042).

Appendix

What each variable recorded

Availability page ► How well do sensors and states cover the dataset?

See the detailed spread in the ▸ Appendix



PL check P-PDG → P-TPT → P-MON-CKP → *PCK open* → P-JUS-CKP

SL check P-MON-CKGL → *CKGL open* → P-JUS-CKGL ← *M2, W2 open* ← P-ANULAR

Valve checks

Open	M1, W1, SDV-P, PCK open; PXO, XO closed
Restart	M1, W1, SDV-P, PCK open
Shut-In, Closed With Diesel or Gas	one of M1, W1, SDV-P, PCK closed
Flushing Diesel or Gas	M1 or W1 closed; PXO or XO open
Depressurization	SDV-P, PCK open; M1 or W1 closed; PXO, XO closed

Plausible ranges

Pressure	Temperature	Valve opening	Flow rate	Valve state
0 to 100 MPa	−50 to 250 °C	0 to 100 %	0 to 10 m^3/s	$<10^8$

What each fault does to its signature

Faults & Features pages ► What does a fault do across sensors, and a sensor across faults?

Fault	Inst.	Wells	From	Topside			Seabed		Subsurface
				ABER-CKP	P-MON-CKP	T-JUS-CKP	P-TPT	T-TPT	P-PDG
Normal Operation	594	9							
Abrupt Increase of BSW	4	3	thesis		↓3/3 (1)	↑3/3 (1)	↓4/4	↑4/4	
Spurious Closure of DHSV	22	7	paper		↓6/7 (15)		↓21/21 (1)	↓22/22	↑15/16 (6)
Severe Slugging	32	2	paper		∼	∼	∼27/32		∼4/31 (1)
Flow Instability	343	7	thesis		∼(112)	∼(195)	∼(85)		∼(309)
Rapid Productivity Loss	11	3	thesis		↓11/11	↓10/11	↓6/10 (1)	↓10/11	
Quick Restriction in PCK	6	2	paper	↓0/1 (5)	↑1/3 (3)		↑4/6		↑-(6)
Scaling in PCK	36	6	thesis		↑36/36	↓27/32 (4)	↑29/31 (5)	↓7/32 (4)	
Hydrate in Production Line	14	9	paper		↓9/14		↑7/8 (5)	↓6/7 (7)	↑9/9 (5)
Hydrate in Service Line	57	15	none		?		?(32)	?(27)	?(28)

↑ ↓ the move the 3W paper [1] or thesis [2] expects from the last hour of normal operation to the last hour of the event

∼ a swing

? no source

a/*b* instances that make it, of those that can show it (Severe Slugging: a swing over the paper's 10 bar).

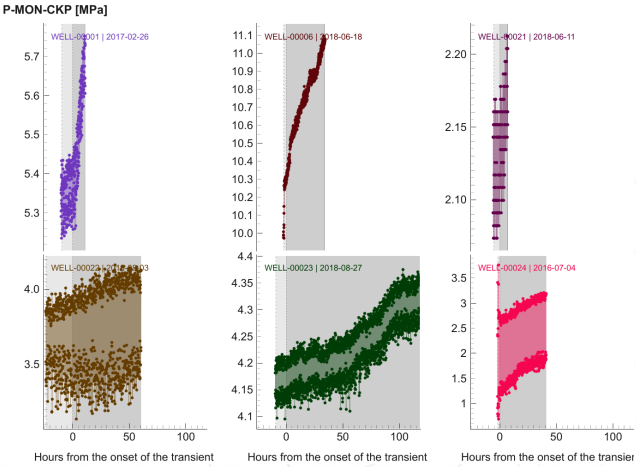
(m) instances where the variable is absent or frozen

Most signatures hold, but each has its exceptions, and some cannot be checked at all

The same fault, well by well

Faults & Features pages ► What does a fault do across sensors, and a sensor across faults?

Scaling in PCK: P-MON-CKP from the onset of the transient, the earliest instance of each well

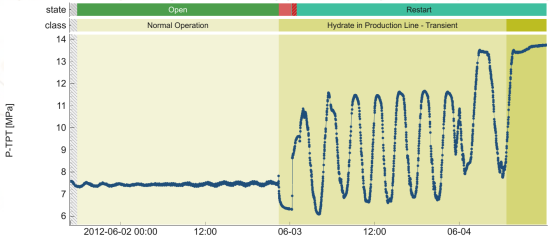


- **Scaling in PCK**: P-MON-CKP rises in 36 of 36, but T-TPT, expected to fall, rises in 15 of WELL-00024's 19 instances.
- **Rapid Productivity Loss**: WELL-00016's P-TPT rises in 4 of 4.
- **Quick Restriction in PCK**: P-PDG is frozen in 6 of 6 instances, the choke opening live in 1.
- **Severe Slugging**: swings P-TPT by 2.6 MPa (median), 27 of 32 over 10 bar, but P-PDG by 0.89 (4 of 31)
- **Flow Instability**: swings P-TPT by 0.07.

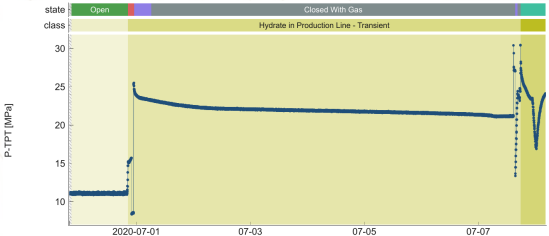
Hydrate in Production Line: the ugly duckling

Faults & Features pages ► What does a fault do across sensors, and a sensor across faults?

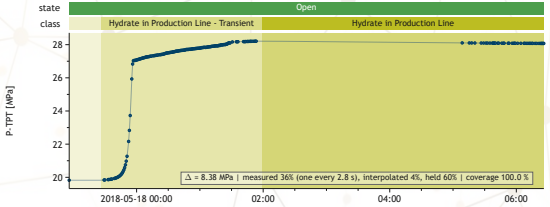
WELL-00019: steady means **plugged**.



WELL-00025: steady means **reopened**.



SIMULATED-00001: steady means **plugged**.



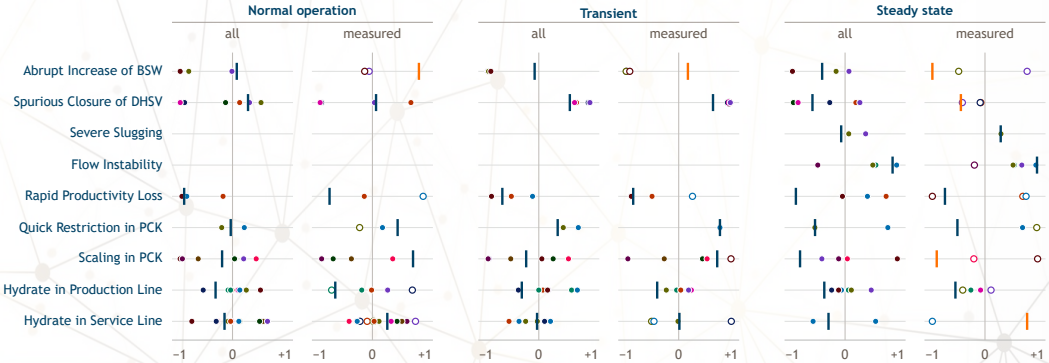
Hydrate labeling across instances, real or simulated, do not seem to share the same philosophy

Appendix

Comparing wells

Dispersions page ► How do sensors correlate across faults?

Each fault's pair ▸ Appendix: a dot per well in its color on the viewer's pages, hollow on fewer than 30 measurements; a bar for the fault's samples pooled, orange on fewer than 30 measurements

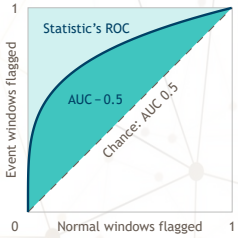


Pooled, a coefficient measures the gaps between wells.

Separating normal from anomalous operation

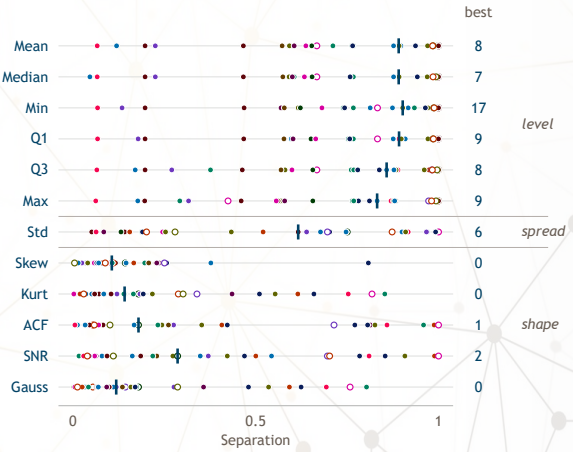
Windows page ► Can statistical features calculated on short horizons represent faults well?

- A binary classifier's separation power is AUC-based:



Separation better than chance
$= 2 \times |\mathrm{AUC} - 0.5|$
$=$ ▬ / (▬ + ▭)
$= 0.6$ here, of AUC 0.8

Statistics' separation ▸ Appendix on windows of 512 samples.



Short windows tell faults by their level, but may indicate when an instance was recorded rather than its fault.

Instances gather by well, not by fault

Instances page ► Is fault behavior consistent across instances and unique to the fault?

Every real instance (1,119) placed as the page places it ▸ Appendix, by its interpolated descriptors. Under each map, how often an instance's nearest neighbour shares its well, its fault, and the nearest of another well its fault (chance: 38%).

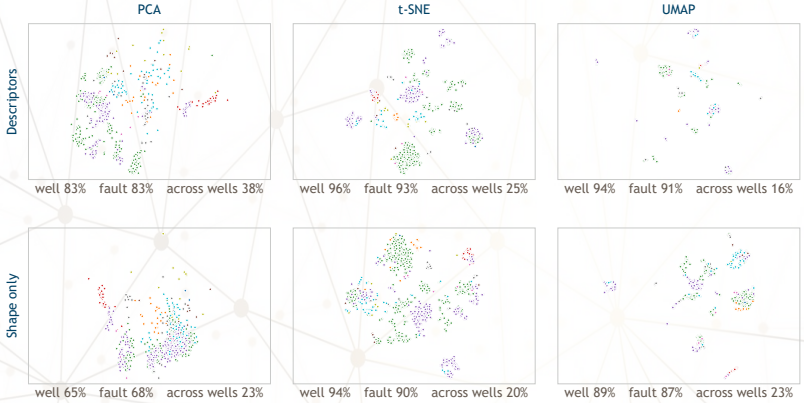


Neighbours share the well before the fault.

The measurements alone change little

Instances page ► Is fault behavior consistent across instances and unique to the fault?

The same maps on the descriptors of the measurements alone, the samples the historian interpolated between them left out.



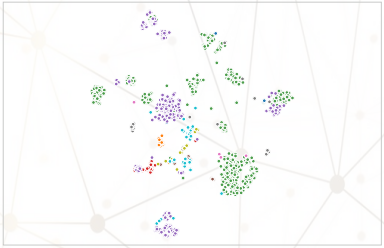
The well still comes first on 4 of the 6 maps.

What knowing the well is worth

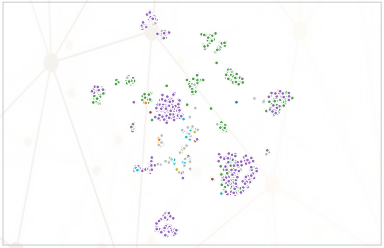
Instances page ► Is fault behavior consistent across instances and unique to the fault?

The map that best gathers the faults (Descriptors, interpolated, t-SNE): every instance in the color of its fault, then in that of the class its well's past names, the one most of the well's earlier instances hold.

by fault



by the well's past



Class		With a past	Named right	
0	Normal Operation	591	432	73%
1	Abrupt Increase of BSW	3	0	0%
2	Spurious Closure of DHSV	15	13	87%
3	Severe Slugging	32	12	38%
4	Flow Instability	341	290	85%
5	Rapid Productivity Loss	10	8	80%
6	Quick Restriction in PCK	6	0	0%
7	Scaling in PCK	32	29	91%
8	Hydrate in Production Line	5	5	100%
9	Hydrate in Service Line	44	36	82%

● a well's first instance

Knowing the well names three instances in four: a model can score by spotting wells.

References

[1] R. E. V. Vargas, A. J. de Melo Júnior, C. B. de Campos Lima et al.
3W Dataset 2.0.0: a realistic and public dataset with rare undesirable real events in oil wells.
Scientific Data 13, 949 (2026).
doi:10.1038/s41597-026-07225-z

[2] R. E. V. Vargas.
Base de dados e benchmarks para prognóstico de anomalias em sistemas de elevação de petróleo.
Doctoral thesis, Universidade Federal do Espírito Santo, Vitória (2019).
http://repositorio.ufes.br/handle/10/13455

PETROBRAS

Thank you!



github.com/Thiago-TP/3w-well-data-viewer

thiagotomasdepaula@gmail.com

Appendix: label changes across an unlabeled stretch

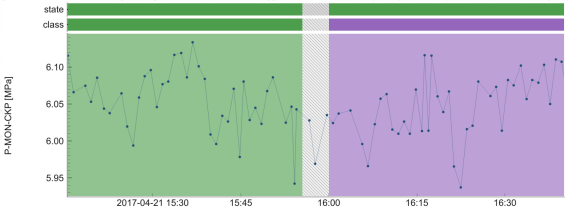
Timelines page ► How are the instances of one well distributed in time?

The 16 unlabeled stretches of a joined history between two different labels.

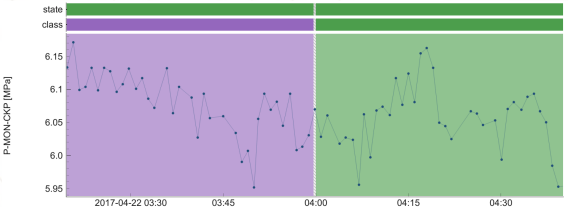
	Well	Unlabeled from	Label before	Label after	For	Shown on
1	WELL-00001	2017-04-21 15:55	normal	Flow Instability (steady)	4 min	P-MON-CKP
2	WELL-00001	2017-04-22 03:59	Flow Instability (steady)	normal	25 s	P-MON-CKP
3	WELL-00001	2017-06-26 09:59	Flow Instability (steady)	normal	3 min	P-MON-CKP
4	WELL-00002	2013-12-12 08:59	normal	Flow Instability (steady)	41 s	P-TPT
5	WELL-00002	2013-12-14 22:00	normal	Flow Instability (steady)	9 s	P-ANULAR
6	WELL-00004	2014-08-06 22:00	Flow Instability (steady)	normal	32 s	P-MON-CKP
7	WELL-00004	2017-10-31 19:14	Quick Restriction in PCK (steady)	normal	5 s	P-MON-CKP
8	WELL-00004	2017-10-31 19:48	Quick Restriction in PCK (steady)	normal	15 s	P-MON-CKP
9	WELL-00006	2018-06-18 11:00	Abrupt Increase of BSW (steady)	normal	30 min	P-MON-CKP
10	WELL-00016	2018-04-26 14:44	Rapid Productivity Loss (transient)	normal	7 min	P-MON-CKP
11	WELL-00020	2014-03-18 01:30	Rapid Productivity Loss (steady)	normal	29 min	P-MON-CKP
12	WELL-00020	2014-03-18 13:57	Rapid Productivity Loss (transient)	normal	53 min	P-MON-CKP
13	WELL-00020	2014-03-19 11:00	Rapid Productivity Loss (transient)	normal	1 h	P-MON-CKP
14	WELL-00024	2016-09-11 07:00	Scaling in PCK (transient)	normal	40 min	P-MON-CKP
15	WELL-00042	2014-12-18 00:53	Hydrate in Service Line (transient)	normal	48 min	P-MON-CKP
16	WELL-00042	2014-12-22 03:37	Hydrate in Service Line (transient)	normal	8 min	P-MON-CKP

Appendix: label changes 1 to 4 (of 16)

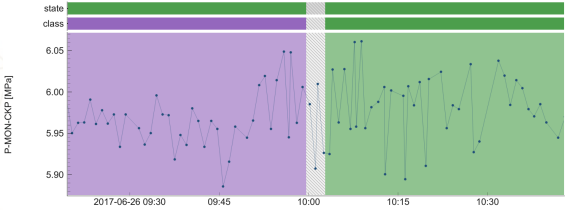
Timelines page ► How are the instances of one well distributed in time?



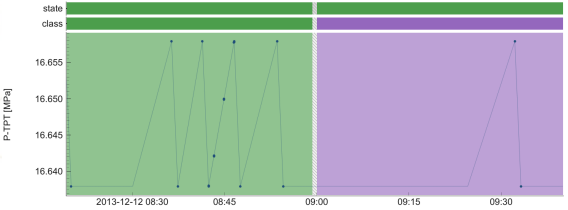
1 WELL-00001, 2017-04-21 15:55: normal → 4 min unlabeled → Flow Instability (steady)



2 WELL-00001, 2017-04-22 03:59: Flow Instability (steady) → 25 s unlabeled → normal



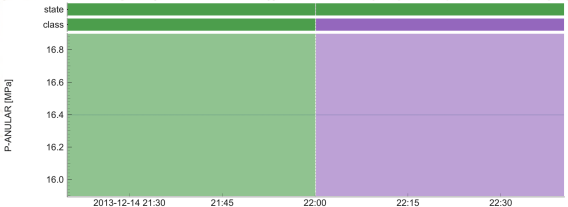
3 WELL-00001, 2017-06-26 09:59: Flow Instability (steady) → 3 min unlabeled → normal



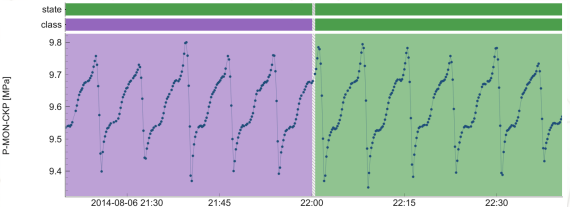
4 WELL-00002, 2013-12-12 08:59: normal → 41 s unlabeled → Flow Instability (steady)

Appendix: label changes 5 to 8 (of 16)

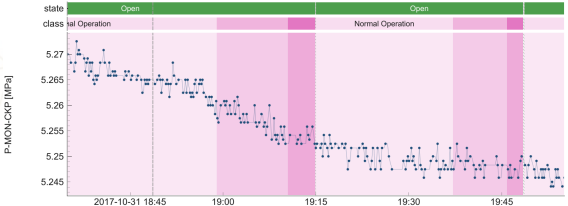
Timelines page ► How are the instances of one well distributed in time?



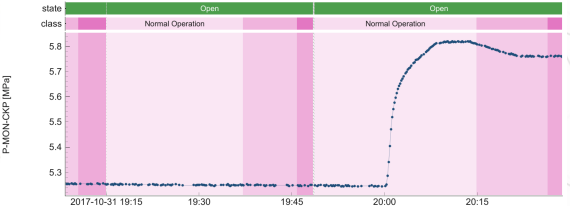
5 WELL-00002, 2013-12-14 22:00: normal → 9 s unlabeled → Flow Instability (steady)



6 WELL-00004, 2014-08-06 22:00: Flow Instability (steady) → 32 s unlabeled → normal



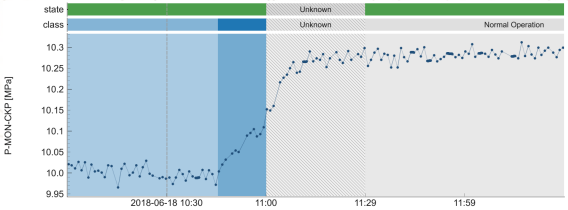
7 WELL-00004, 2017-10-31 19:14: Quick Restriction in PCK (steady) → 5 s unlabeled → normal



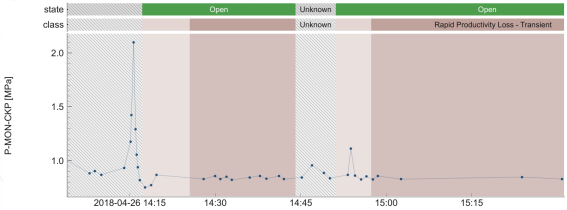
8 WELL-00004, 2017-10-31 19:48: Quick Restriction in PCK (steady) → 15 s unlabeled → normal

Appendix: label changes 9 to 12 (of 16)

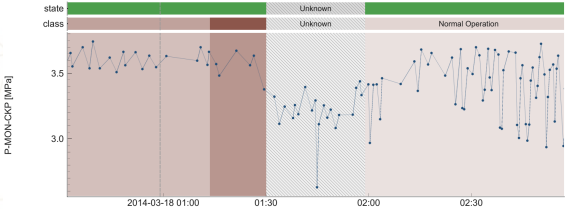
Timelines page ► How are the instances of one well distributed in time?



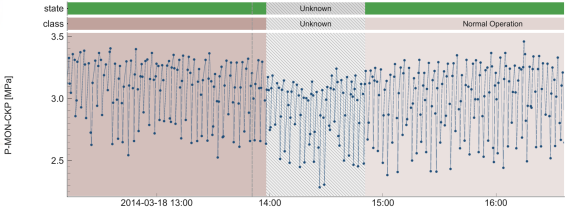
9 WELL-00006, 2018-06-18 11:00: Abrupt Increase of BSW (steady) → 30 min unlabeled → normal



10 WELL-00016, 2018-04-26 14:44: Rapid Productivity Loss (transient) → 7 min unlabeled → normal



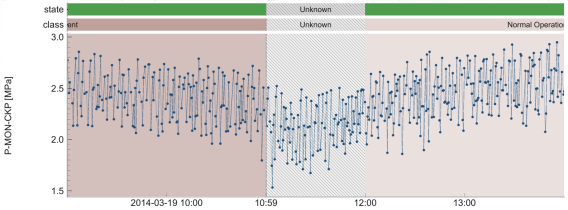
11 WELL-00020, 2014-03-18 01:30: Rapid Productivity Loss (steady) → 29 min unlabeled → normal



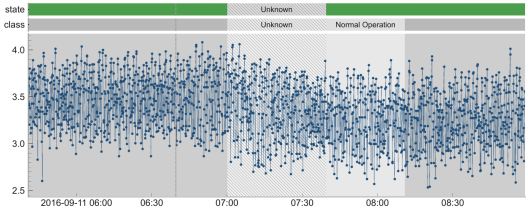
12 WELL-00020, 2014-03-18 13:57: Rapid Productivity Loss (transient) → 53 min unlabeled → normal

Appendix: label changes 13 to 16 (of 16)

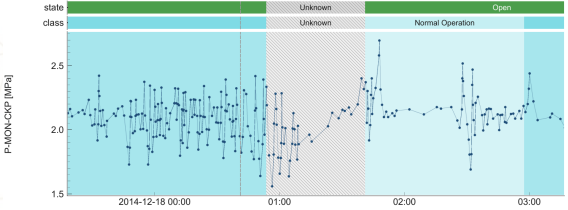
Timelines page ► How are the instances of one well distributed in time?



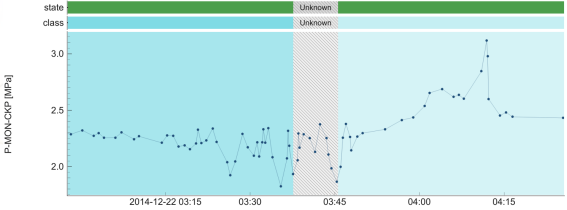
13 WELL-00020, 2014-03-19 11:00: Rapid Productivity Loss (transient) → 1 h unlabeled → normal



14 WELL-00024, 2016-09-11 07:00: Scaling in PCK (transient) → 40 min unlabeled → normal



15 WELL-00042, 2014-12-18 00:53: Hydrate in Service Line (transient) → 48 min unlabeled → normal



16 WELL-00042, 2014-12-22 03:37: Hydrate in Service Line (transient) → 8 min unlabeled → normal

Appendix: what each variable recorded

Availability page ► How well do sensors and states cover the dataset?

Variable	Live			Frozen			Implausible			Out of order			Against state			Missing			Farthest out
	samples	%	inst.	samples	%	inst.	samples	%	inst.	samples	%	inst.	samples	%	inst.	samples	%	inst.	
T-TPT	26,178,605	79.6	1,006	2,236,395	6.8	29	0	0.0	0	-	-	-	-	-	-	4,457,250	13.6	84	
P-TPT	25,878,401	78.7	922	1,612,066	4.9	38	1,004,849	3.1	10	361,705	1.1	29	-	-	-	5,381,783	16.4	159	2942 MPa
P-MON-CKP	25,749,471	78.3	720	42,928	0.1	2	12,489	0.0	5	104,739	0.3	6	-	-	-	7,079,851	21.5	397	−0.237 MPa
P-ANULAR	25,320,751	77.0	914	139,764	0.4	7	387,650	1.2	3	1,296,207	3.9	6	-	-	-	7,411,735	22.5	198	−7.658 MPa
ESTADO-SDV-P	24,316,107	74.0	661	0	0.0	0	0	0.0	0	-	-	-	653,430	2.0	23	8,556,143	26.0	458	
P-JUS-CKGL	24,307,254	73.9	656	743,125	2.3	33	1,299,132	4.0	85	1,298,303	4.0	7	-	-	-	7,821,871	23.8	430	−0.4977 MPa
ESTADO-W1	18,938,036	57.6	558	0	0.0	0	0	0.0	0	-	-	-	3,086,682	9.4	14	13,934,214	42.4	561	
ESTADO-XO	18,538,730	56.4	533	0	0.0	0	0	0.0	0	-	-	-	3,351	0.0	1	14,333,520	43.6	586	
ESTADO-W2	18,293,462	55.7	533	0	0.0	0	0	0.0	0	-	-	-	-	-	-	14,578,788	44.4	586	
ESTADO-PXO	18,176,203	55.3	532	0	0.0	0	0	0.0	0	-	-	-	57,977	0.2	3	14,696,047	44.7	587	
ESTADO-M1	17,773,205	54.1	501	0	0.0	0	0	0.0	0	-	-	-	8,889	0.0	3	15,099,045	45.9	618	
ESTADO-M2	17,688,282	53.8	504	0	0.0	0	0	0.0	0	-	-	-	-	-	-	15,183,968	46.2	615	
ESTADO-SDV-GL	17,105,271	52.0	480	0	0.0	0	0	0.0	0	-	-	-	-	-	-	15,766,979	48.0	639	
T-JUS-CKP	16,689,089	50.8	554	150,261	0.5	8	0	0.0	0	-	-	-	-	-	-	16,032,900	48.8	557	
ESTADO-DHSV	13,865,397	42.2	470	0	0.0	0	0	0.0	0	-	-	-	-	-	-	19,006,853	57.8	649	
QGL	11,150,903	33.9	147	6,569,202	20.0	298	0	0.0	0	-	-	-	-	-	-	15,152,145	46.1	674	
ABER-CKP	9,248,257	28.1	272	3,078,201	9.4	27	161,872	0.5	2	-	-	-	11,955	0.0	4	20,545,792	62.5	820	−99.99 %
P-PDG	7,119,658	21.7	174	17,836,376	54.3	653	1,200,682	3.7	33	256,966	0.8	24	-	-	-	7,916,216	24.1	292	-1.2×10^{36} MPa
T-PDG	6,426,251	19.5	142	14,385,021	43.8	421	2,445,015	7.4	36	-	-	-	-	-	-	12,060,978	36.7	556	-1.7×10^{38} °C
T-MON-CKP	4,789,008	14.6	14	161,538	0.5	1	161,538	0.5	1	-	-	-	-	-	-	27,921,704	84.9	1,104	−99.99 °C
P-JUS-CKP	3,287,363	10.0	10	161,538	0.5	1	1,506,941	4.6	7	0	0.0	0	-	-	-	29,423,349	89.5	1,108	1559 MPa
ABER-CKGL	1,403,126	4.3	48	7,088,387	21.6	238	0	0.0	0	-	-	-	-	-	-	24,380,737	74.2	833	
P-MON-CKGL	308,726	0.9	2	0	0.0	0	6,401	0.0	1	2,096	0.0	1	-	-	-	32,563,524	99.1	1,117	−0.094 MPa
P-JUS-BS	0	0.0	0	0	0.0	0	0	0.0	0	-	-	-	-	-	-	32,872,250	100.0	1,119	
P-MON-SDV-P	0	0.0	0	0	0.0	0	0	0.0	0	-	-	-	-	-	-	32,872,250	100.0	1,119	
PT-P	0	0.0	0	0	0.0	0	0	0.0	0	-	-	-	-	-	-	32,872,250	100.0	1,119	
QBS	0	0.0	0	0	0.0	0	0	0.0	0	-	-	-	-	-	-	32,872,250	100.0	1,119	

Appendix: hydrate instances transients and steadies

Faults & Features pages ► What does a fault do across sensors, and a sensor across faults?

Well	From	Transient			Steady		
		h	state	%	h	state	%
WELL-00019	2012-06-01	32	Restart	94	6	Restart	100
WELL-00019	2014-11-17	73	Shut-In	98	2	Flushing Diesel	100
WELL-00019	2015-06-14	30	Shut-In	92	1	Bullheading	100
WELL-00019	2021-06-17	53	Shut-In	86	2	Flushing Diesel	100
WELL-00025	2020-06-29	165	Closed With Gas	93	11	Restart	100
WELL-00026	2016-06-01	43	Open	100	2	Open	100
WELL-00026	2017-02-07	126	Open	100		never	
WELL-00026	2017-06-08	79	Open	100		never	
WELL-00027	2023-09-18	166	Shut-In	100	8	Flushing Diesel	100
WELL-00028	2021-06-17	76	Shut-In	65	2	Restart	100
WELL-00029	2020-10-20	23	Shut-In	72	1	Bullheading	100
WELL-00030	2014-11-08	33	Shut-In	81	2	Flushing Diesel	100
WELL-00031	2014-04-03	92	Shut-In	90	1	Bullheading	100
WELL-00032	2011-08-30	24	Shut-In	98	2	Flushing Diesel	100
81 simulated			Open	100		Open	100

Appendix: every fault's coefficients

Dispersions page ► How do sensors correlate across faults?

Fault	Pair	Normal operation			Transient			Steady state		
		all	measured		all	measured		all	measured	
Abrupt Increase of BSW	P-TPT × T-TPT	+0.08	+0.88	(24)	−0.08	+0.17	(20)	−0.44	−1.00	(6)
Spurious Closure of DHSV	P-TPT† × T-TPT	+0.29	+0.07	(314)	+0.59	+0.65	(494)	−0.62	−0.46	(28)
Severe Slugging	P-TPT × T-JUS-CKP	-	-		-	-		−0.07	+0.30	(1,348)
Flow Instability	P-MON-CKP† × P-TPT†	-	-		-	-		+0.90	+0.99	(11,999)
Rapid Productivity Loss	P-ANULAR × T-TPT	−0.92	−0.82	(366)	−0.69	−0.86	(394)	−0.94	−0.76	(42)
Quick Restriction in PCK	P-ANULAR × P-TPT	−0.03	+0.48	(1,132)	+0.36	+0.78	(420)	−0.57	−0.52	(130)
Scaling in PCK	P-JUS-CKGL × P-MON-CKP	−0.20	+0.77	(1,555)	−0.24	+0.73	(12,891)	−0.86	−0.92	(16)
Hydrate in Production Line	P-MON-CKP × QGL	−0.32	−0.71	(7,349)	−0.32	−0.41	(22,644)	−0.40	−0.56	(493)
Hydrate in Service Line	P-JUS-CKGL × P-MON-CKP	−0.15	+0.28	(14,053)	−0.03	+0.01	(3,125)	−0.32	+0.80	(15)

Pair of best covering variables available in all instances of a fault

† instances of a fault don't share a pair and defaulted to the best covered of its signature [1, 2]

all Pearson's coefficient over every plausible sample of the period, pooled over the fault's instances

measured over the samples at which both were measured, their count in brackets, set apart below 30

Appendix: every well's coefficients (1 of 3)

Dispersions page ► How do sensors correlate across faults?

Fault	Pair	Well	Normal operation		Transient		Steady state	
			all	measured	all	measured	all	measured
Abrupt Increase of BSW	P-TPT × T-TPT	WELL-00001	−0.01	−0.06 (5)	−0.97	−0.99 (6)	+0.07	+0.80 (3)
		WELL-00002	−0.83	-	−0.96	−1.00 (4)	−0.17	−0.50 (3)
		WELL-00006	−0.99	−0.15 (18)	−0.91	−0.93 (10)	−1.00	-
Spurious Closure of DHSV	P-TPT† × T-TPT	WELL-00002	+0.54	-	+0.97	+0.96 (6)	-	-
		WELL-00003	+0.14	+0.73 (95)	+0.72	+0.93 (51)	+0.20	−0.08 (3)
		WELL-00009	−0.13	-	+0.95	+0.94 (11)	−0.98	-
		WELL-00011	−0.91	−0.96 (115)	+0.94	+0.92 (287)	−0.29	−0.09 (17)
		WELL-00012	−0.99	−0.99 (53)	+0.68	+0.97 (4)	−0.89	-
		WELL-00013	+0.32	+0.04 (51)	+0.97	+0.98 (135)	+0.28	−0.43 (5)
Severe Slugging	P-TPT × T-JUS-CKP	WELL-00001	-	-	-	-	+0.39	-
		WELL-00014	-	-	-	-	+0.07	+0.30 (1,347)
Flow Instability	P-MON-CKP† × P-TPT†	WELL-00001	-	-	-	-	+0.57	+0.70 (367)
		WELL-00004	-	-	-	-	+0.98	+0.97 (126)
		WELL-00005	-	-	-	-	+0.59	+0.56 (6,253)
		WELL-00007	-	-	-	-	−0.52	−0.20 (12)
		WELL-00014	-	-	-	-	+0.53	+0.53 (5,241)
Rapid Productivity Loss	P-ANULAR × T-TPT	WELL-00015	−0.18	−0.15 (360)	−0.52	−0.51 (272)	+0.78	+0.72 (25)
		WELL-00016	−0.87	+0.96 (5)	−0.12	+0.26 (4)	+0.42	+0.78 (15)
		WELL-00020	−0.96	-	−0.90	−0.90 (118)	−0.05	−1.00 (2)

Appendix: every well's coefficients (2 of 3)

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Dispersions page ► How do sensors correlate across faults?

Fault	Pair	Well	Normal operation			Transient			Steady state		
			all	measured		all	measured		all	measured	
Quick Restriction in PCK	P-ANULAR × P-TPT	WELL-00002	−0.20	−0.24	(6)	+0.46	-		−0.57	+0.98	(7)
		WELL-00004	+0.22	+0.19	(1,126)	+0.75	+0.78	(420)	+0.81	+0.71	(123)
Scaling in PCK	P-JUS-CKGL × P-MON-CKP	WELL-00001	+0.21	-		−0.95	-		−0.44	-	
		WELL-00006	−1.00	-		+0.06	+0.99	(4)	+0.99	+1.00	(2)
		WELL-00021	−0.95	−0.96	(89)	−0.96	−0.97	(77)	−0.12	-	
		WELL-00022	−0.65	−0.40	(256)	−0.53	−0.27	(1,769)	-	-	
		WELL-00023	+0.04	−0.75	(111)	+0.27	+0.45	(1,245)	-	-	
		WELL-00024	+0.45	+0.39	(1,099)	+0.56	+0.54	(9,796)	+0.04	−0.21	(13)
Hydrate in Production Line	P-MON-CKP × QGL	WELL-00019	−0.08	−0.20	(1,687)	+0.62	−0.04	(1,524)	+0.06	−0.27	(299)
		WELL-00025	−0.56	+0.76	(18)	−0.39	-		−0.26	-	
		WELL-00026	+0.10	+0.28	(4,869)	+0.11	+0.24	(19,618)	−0.10	−0.08	(183)
		WELL-00027	+0.25	+0.29	(113)	+0.18	+0.19	(216)	+0.50	+0.12	(5)
		WELL-00028	+0.26	-		+0.12	−0.24	(1,176)	+0.12	−0.42	(5)
		WELL-00029	−0.01	−0.02	(654)	+0.08	+0.04	(108)	-	-	
		WELL-00030	+0.14	-		+0.73	-		−0.07	-	
		WELL-00031	−0.04	−0.77	(8)	−0.00	-		-	-	
		WELL-00032	+0.53	-		+0.17	-		−0.12	-	

Appendix: every well's coefficients (3 of 3)

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Dispersions page ► How do sensors correlate across faults?

Fault	Pair	Well	Normal operation		Transient		Steady state	
			all	measured	all	measured	all	measured
Hydrate in Service Line	P-JUS-CKGL × P-MON-CKP	WELL-00010	−0.28	−0.30 (111)	-	-	-	-
		WELL-00014	+0.14	+0.13 (2,330)	−0.03	−0.01 (3,085)	-	-
		WELL-00015	−0.03	−0.10 (6)	-	-	-	-
		WELL-00016	+0.59	+0.82 (33)	+0.22	-	+0.58	-
		WELL-00020	−0.77	-	-	-	-	-
		WELL-00033	+0.66	+0.66 (2,285)	-	-	-	-
		WELL-00034	+0.58	+0.56 (2,518)	-	-	-	-
		WELL-00035	+0.51	+0.47 (3,998)	-	-	-	-
		WELL-00036	−0.30	−0.44 (1,560)	-	-	-	-
		WELL-00037	−0.31	−0.24 (21)	+0.11	+1.00 (2)	-	-
		WELL-00038	−0.02	+0.36 (37)	-	-	-	-
		WELL-00039	+0.67	+0.82 (10)	-	-	-	-
		WELL-00040	−0.09	+0.02 (20)	−0.25	−0.52 (8)	-	-
		WELL-00041	−0.03	+0.03 (1,092)	−0.57	-	-	-
		WELL-00042	+0.12	−0.29 (32)	−0.38	−0.47 (28)	−0.61	−1.00 (2)

Appendix: every statistic, well by well (1 of 2)

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Windows page ► Can statistical features calculated on short horizons represent faults well?

Each well's best covered analog sensor (most samples live; ties, Tied, to the one separating best). A well's best separation at 512 samples in bold, its line orange on fewer than 30 windows of a kind; across window lengths it moves by 0.07 at most.

Well	Sensor	Tied	Windows		Level						Spread	Shape				
			normal	event	Mean	Median	Min	Q1	Q3	Max	Std	Skew	Kurt	ACF	SNR	Gauss
WELL-00001	P-TPT	4	3,454	680	0.23	0.23	0.13	0.18	0.27	0.32	**0.64**	0.03	0.12	0.28	0.37	0.12
WELL-00002	T-TPT	1	7,291	1,683	0.59	0.59	**0.62**	0.60	0.58	0.57	0.43	0.02	0.01	0.24	0.27	0.00
WELL-00003	P-MON-CKP	1	949	40	0.97	0.97	0.97	0.97	0.97	0.97	0.52	0.23	0.62	0.99	**0.99**	0.59
WELL-00004	P-MON-CKP	1	181	629	0.12	0.05	0.81	0.60	0.17	0.18	**0.90**	0.38	0.06	0.00	0.08	0.11
WELL-00005	P-TPT	4	1,942	542	0.71	0.60	0.93	0.88	0.38	0.29	**0.97**	0.20	0.85	0.96	0.91	0.80
WELL-00006	T-TPT	1	4,243	541	0.20	0.20	0.20	**0.20**	0.20	0.20	0.05	0.12	0.10	0.08	0.09	0.10
WELL-00007	P-JUS-CKGL	6	30	147	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	0.92	0.23	0.44	0.81	0.78	0.36
WELL-00008	P-JUS-CKGL	1	1,994	0	-	-	-	-	-	-	-	-	-	-	-	-
WELL-00009	P-ANULAR	6	7	8	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	0.75	0.14	0.18	0.18	0.29	0.18
WELL-00010	T-TPT	1	419	1,192	0.89	0.89	0.90	0.89	0.89	0.87	**0.92**	0.25	0.75	0.86	0.81	0.69
WELL-00011	P-TPT	2	91	140	0.94	0.94	**0.99**	0.97	0.84	0.78	0.70	0.26	0.51	0.77	0.78	0.48
WELL-00012	P-ANULAR	7	22	6	0.67	0.67	0.83	0.83	0.67	0.42	**1.00**	0.11	0.82	**1.00**	**1.00**	0.76
WELL-00013	P-ANULAR	3	8	14	**1.00**	**1.00**	**1.00**	**1.00**	0.98	0.97	0.70	0.25	0.34	0.71	0.70	0.14
WELL-00014	QGL	1	99	1,530	0.97	0.97	**0.97**	0.97	0.96	0.88	0.91	0.11	0.55	0.83	0.91	0.53
WELL-00015	ABER-CKP	8	41	20	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	0.87	0.11	0.29	0.05	0.70	0.05
WELL-00016	T-TPT	9	120	56	0.90	0.89	**0.92**	0.91	0.86	0.83	0.20	0.15	0.04	0.01	0.16	0.03
WELL-00019	QGL	2	757	1,407	0.76	0.76	0.74	0.75	0.76	**0.76**	0.75	0.11	0.17	0.05	0.02	0.17
WELL-00020	T-TPT	2	110	603	0.47	0.47	**0.47**	0.47	0.46	0.46	0.14	0.08	0.07	0.08	0.23	0.06
WELL-00021	P-MON-CKP	6	88	87	**0.60**	0.60	0.57	0.60	0.60	0.58	0.08	0.23	0.08	0.07	0.06	0.09

Appendix: every statistic, well by well (2 of 2)

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Windows page ► Can statistical features calculated on short horizons represent faults well?

Well	Sensor	Tied	Windows		Level						Spread	Shape				
			normal	event	Mean	Median	Min	Q1	Q3	Max	Std	Skew	Kurt	ACF	SNR	Gauss
WELL-00022	T-JUS-CKP	1	844	5,169	0.57	0.57	**0.58**	0.58	0.57	0.56	0.20	0.21	0.14	0.26	0.23	0.13
WELL-00023	P-MON-CKP	1	361	2,664	0.65	0.65	0.62	0.65	**0.66**	0.66	0.13	0.01	0.02	0.10	0.11	0.02
WELL-00024	P-JUS-CKGL	1	644	4,675	0.07	0.07	**0.07**	0.07	0.07	0.06	0.06	0.01	0.00	0.00	0.03	0.01
WELL-00025	P-JUS-CKP	2	168	1,237	0.99	0.99	0.96	0.99	0.99	0.99	**0.99**	0.81	0.66	0.81	0.85	0.62
WELL-00026	T-MON-CKP	8	754	1,758	0.64	0.63	**0.68**	0.66	0.60	0.56	0.24	0.06	0.00	0.07	0.06	0.01
WELL-00027	P-MON-CKP	5	96	1,227	1.00	1.00	**1.00**	**1.00**	1.00	1.00	0.97	0.02	0.18	0.04	0.04	0.28
WELL-00028	T-MON-CKP	1	86	545	0.89	0.89	**0.90**	0.90	0.88	0.88	0.25	0.04	0.22	0.35	0.42	0.16
WELL-00029	P-JUS-CKP	8	112	169	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	0.99	0.15	0.17	0.02	0.41	0.50	0.03
WELL-00030	ABER-CKGL	10	62	248	**1.00**	**1.00**	1.00	**1.00**	**1.00**	**1.00**	0.75	0.01	0.05	0.17	0.35	0.01
WELL-00031	T-MON-CKP	2	51	651	0.77	0.77	0.77	0.77	0.77	**0.77**	0.19	0.07	0.14	0.23	0.22	0.04
WELL-00032	P-TPT	5	71	185	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	0.62	0.00	0.06	0.04	0.19	0.07
WELL-00033	P-JUS-CKGL	3	291	0	-	-	-	-	-	-	-	-	-	-	-	-
WELL-00034	P-JUS-CKGL	3	365	0	-	-	-	-	-	-	-	-	-	-	-	-
WELL-00035	P-JUS-CKGL	3	510	0	-	-	-	-	-	-	-	-	-	-	-	-
WELL-00036	P-JUS-CKGL	4	35	0	-	-	-	-	-	-	-	-	-	-	-	-
WELL-00037	P-JUS-CKGL	7	129	30	0.77	0.76	0.74	0.76	0.78	**0.81**	0.19	0.07	0.19	0.42	0.47	0.17
WELL-00038	ABER-CKP	9	236	0	-	-	-	-	-	-	-	-	-	-	-	-
WELL-00039	P-ANULAR	5	116	0	-	-	-	-	-	-	-	-	-	-	-	-
WELL-00040	P-ANULAR	5	35	26	0.98	0.99	0.99	0.98	**1.00**	0.99	0.28	0.00	0.30	0.10	0.11	0.29
WELL-00041	P-PDG	2	1,893	23	0.98	0.98	**0.99**	0.99	0.98	0.98	0.20	0.09	0.03	0.06	0.04	0.01
WELL-00042	P-JUS-CKGL	9	99	190	0.88	0.88	0.87	0.88	0.88	**0.88**	0.68	0.07	0.08	0.03	0.54	0.07

Appendix: what places an instance on the maps

PETROBRAS

Instances page ▶ Is fault behavior consistent across instances and unique to the fault?

Per instance, the 5 sensors live in at least 50% of the instances (P-ANULAR, P-JUS-CKGL, P-MON-CKP, P-TPT and T-TPT), each by the descriptors below: all of them (Descriptors, 60 columns) or those that leave the level out (Shape only, 35). They are taken on the instance's labeled samples, over the 1 Hz grid (interpolated) or over the samples the historian measured (measurements), the sampling ones alike. Every column is standardized, and a cell an instance lacks (its sensor absent or frozen) takes the column's median.

Descriptor	Reads	Descriptors	Shape only
Mean	level	✓	
Standard deviation	level	✓	
Median	level	✓	
5th percentile	level	✓	
95th percentile	level	✓	
Skewness	shape	✓	✓
Kurtosis	shape	✓	✓
Autocorrelation time (log)	shape	✓	✓
Signal-to-noise ratio (log)	shape	✓	✓
Gaussianity slope	shape	✓	✓
Share of samples measured	sampling	✓	✓
Spacing of the measurements (log)	sampling	✓	✓

Chance. Two different instances drawn at random, the second from those left, share a fault class with probability

$$\sum_{c=0}^{9} \frac{n_c}{N}\frac{n_c-1}{N-1} = \frac{594\cdot 593 + 4\cdot 3 + \cdots + 57\cdot 56}{1{,}119\cdot 1{,}118}$$

$$= \frac{475{,}788}{1{,}251{,}042} = 0.380,$$

n_c the instances of class c and N all of them.

Appendix: how the maps lay the instances out

Instances page ► Is fault behavior consistent across instances and unique to the fault?

PCA: the two principal components of the standardized columns (numpy's SVD), which carry the share of the variance below.
t-SNE: scikit-learn's, perplexity 30, initialized by PCA, Euclidean distances, seed 0.
UMAP: umap-learn's, 15 neighbours, Euclidean distances, seed 0, its other settings the defaults.

Representation	Data	Columns		PCA	Neighbours kept		
		count	imputed	variance	PCA	t-SNE	UMAP
Descriptors	interpolated	60	26%	18% + 15%	35%	67%	57%
Descriptors	measurements	60	28%	16% + 14%	32%	66%	56%
Shape only	interpolated	35	26%	21% + 9%	23%	62%	55%
Shape only	measurements	35	31%	16% + 12%	22%	61%	53%

Neighbours kept of an instance's 10 nearest neighbours in the columns, the share still among its 10 nearest on the map.